

Answer Key & Teacher Notes

Session 1 – Integers and Opposites • Session 2 – Integers in Real Situations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Integers and Opposites

TALK PROMPTS

- How does a number line show that two opposite elevations are the same distance from zero?
- Why do adding money and using money up get opposite signs?

PARTNER TASK

Elevator Trip. One partner writes integers for 5 floors down and 5 floors up from the lobby; the other explains what 0 means. Switch roles with Tower and Tunnel: 44 feet above and 44 feet below the ground.

WATCH FOR

- Dropping the negative sign for “below” or “debt” (writing 24 for 24 m below sea level).
- Thinking a negative number is its own opposite (opposite of -18 is -18).
- Writing an integer before deciding what 0 means in the situation.

SESSION 2 • TEACHER NOTES

Integers in Real Situations

TALK PROMPTS

- Two cities can sit the same distance from zero. Why are their temperatures still different?
- How do the two signs show that a 45-dollar refund and a 45-dollar charge cancel each other out?

PARTNER TASK

Refunds and Yards. One partner writes integers for a 45-dollar charge and a 45-dollar refund; the other writes integers for a 9-yard loss and a 9-yard gain. Each explains what a total of 0 means.

WATCH FOR

- Believing two opposites always make a positive (opposite of the opposite of -19 written as 19).
- Using a positive number for a loss, drop, or withdrawal.
- Thinking two readings the same distance from 0 are equal.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.C.5	1	1	2	1	2	2	2	3	3	1	1
S1 • Form B	6.NS.C.5	1	1	2	1	2	2	2	3	3	1	1
S2 • Form A	6.NS.C.5	1	1	2	1	1	2	2	3	3	1	1
S2 • Form B	6.NS.C.5	1	1	2	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

integer: Zero, a counting number, or the **opposite** of a counting number.

opposite: A number the same **distance** from 0, on the other side of 0.

negative number: A number **less** than 0, written with a $-$ sign.

positive number: A number **greater** than 0.

- First decide what **0** means in the situation.
- Gains, above, and deposits are **positive**. Losses, below, and debts are **negative**.

Example: Zero means the **starting** score. Earning 8 points is **8**. Losing 8 points is **-8**. Each is 8 units from 0, on opposite sides. Answer: 8 and -8 are **opposites**.

YOU TRY

1. a. -14 b. 6
2. Points at -4 and 4

EXIT TICKET

1. a. -20 b. 11
2. Points at -5 and 5; 5 units each
3. **A** (-4 and 4)

PRACTICE FORM A

1. a. 12 b. -25 c. -8 d. -3
2. a. 19 b. -6 c. 0
3. a. 35, -20, 15, -35 b. Ridge Point and Cave Floor (35 and -35)
4. -12; opposite 12
5. No; the opposite of -18 is 18
6. -7, 7, 0; opposite pair: -7 and 7
7. a. -31 ft (windmill 31) b. ground level
8. a. -3 and 7 b. 4 c. -4 means 4 floors below the lobby
9. Below sea level is negative, so the elevation is -24 m. An opposite is on the other side of 0, so the opposite of -24 is 24 (24 m above sea level).
10. **B** (-30)
11. **C** (0)

Self-check word INTEGER

PRACTICE FORM B

1. a. 17 b. -32 c. -6 d. -4
2. a. 23 b. -9 c. 0
3. a. 28, -16, 12, -28 b. Lookout and Dive Station (28 and -28)
4. -15; opposite 15
5. No; the opposite of -21 is 21
6. -5, 5, 0; opposite pair: -5 and 5
7. a. -44 ft (tower 44) b. ground level
8. a. -5 and 8 b. 3 c. -3 means 3 floors below the lobby
9. Below sea level is negative, so the elevation is -26 m. An opposite is on the other side of 0, so the opposite of -26 is 26 (26 m above sea level).
10. **B** (9)
11. **C** (15)

Self-check word NEGATIVE

GUIDED NOTES

opposite: A number the same distance from 0, on the **other** side of 0.

opposite of the opposite: Flip across 0 **twice** and you are back where you started.

integer: A whole number or its **opposite**.

- The sign shows the **direction** of a change from a starting point.
- $-(-9)$ means the opposite of -9 , which is **9**.

Example: Zero means **no change** in the water. Monday's gain is **25**. Tuesday's loss is **-25**. Answer: The two changes are **opposites**: same size, opposite effects.

YOU TRY

1. a. 11 b. 9
2. -14 m (first flip 14, second flip -14)

EXIT TICKET

1. a. -20 b. 13
2. -7 ; opposite 7, which means 7 degrees above zero
3. **B** (9)

PRACTICE FORM A

1. a. -9 b. 45 c. -6 d. 200
2. a. 16 b. -7 c. 21 d. -8
3. a. points at -15 , -8 , 8, 21 b. Denver and Oslo c. 8 units
4. 33 degrees
5. -7 and 7 (0 is ground level)
6. No; -26 goes to 26, then back to -26 . The value is -26 .
7. -30 , -11 , 0, 30; opposite pair: -30 and 30
8. a. 36 m b. -36 m c. the cave entrance
9. The first flip takes -19 to 19. The second flip goes back across 0 to -19 . Two opposites return to the start, so the answer is -19 .
10. **B** (45)
11. **B** (-75)

Self-check word **OPPOSITE**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • INTEGERS AND OPPOSITES

1. a. -12 b. 40 c. -9 d. 15 e. -25 f. 7
2. Points at -6 and 6; each is 6 units from 0.
3. A: -7 , opposite 7; B: -2 , opposite 2; C: 3, opposite -3 ; D: 9, opposite -9
4. Opposites are on different sides of 0, so the sign changes. -15 is 15 units left of 0; its opposite is 15 units right of 0. The opposite of -15 is 15.

PRACTICE FORM B

1. a. -12 b. 30 c. -4 d. 150
2. a. 24 b. -5 c. 38 d. -6
3. a. points at -13 , -6 , 6, 19 b. Pine Lodge and Summit Lodge c. 6 units
4. 27 degrees
5. -9 and 9 (0 is ground level)
6. No; -14 goes to 14, then back to -14 . The value is -14 .
7. -25 , -8 , 0, 25; opposite pair: -25 and 25
8. a. 42 m b. -42 m c. the cliff top
9. The second opposite must also flip across 0: 22 goes to -22 . Two opposites return to the start, so the answer is -22 .
10. **B** (52)
11. **B** (-60)

Self-check word **POSITIVE**

SHEET 2 • INTEGERS IN REAL SITUATIONS

1. a. -35 b. 20 c. -4 d. -8
2. a. 9 b. -15 c. -4
3. A at -3 , B at 3, C at -3 . A and C are the same point: the opposite of the opposite of -3 is -3 .
4. a. noon 6, midnight -6 b. They are opposites: each is 6 units from 0, on different sides.
5. No. The opposite of a negative number is positive. Example: the opposite of -7 is 7, which is positive.
6. **B** (-18)
7. **B** ($-(-11) = 11$)